Lecture 9: Improper Integrals

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Know what is meant for an improper integral with unbounded domain to converge or diverge.
- ☐ Know what the values of p such that the p-integrals converge or diverge.
- ☐ Know what is meant for an improper integral involving integrand with vertical asymptote to converge or diverge.
- ☐ Use the direct comparison test to tell whether an improper integral converges or diverges when possible and know the limitations of the theorem.
- ☐ Use the limit comparison test to tell whether an improper integral converges or diverges when possible and know the limitations of the theorem.
- ☐ Identify the p-integral to be used for the comparison test even if the question does not suggest it.

Introduction

So far, we have considered definite integrals of the form $I = \int_a^b f(x)\,dx$, where the integrand f is continuous over the interval $[a, b]$.

Such integral I is necessarily a finite number and is called proper integrals.

Let $I = \int_a^b f(x)\,dx$ be a definite integral,

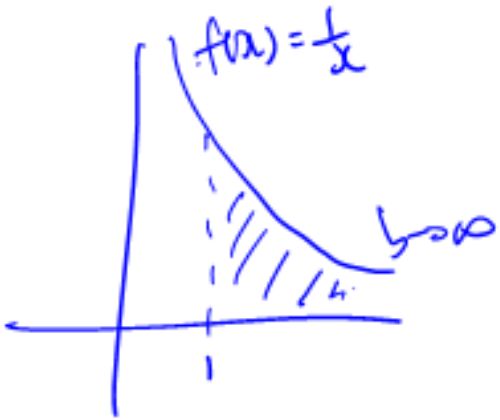
I is proper when f is cts over $[a,b]$ and $[a,b]$ is a finite interval.
] is finite Eg: $\int_0^2 3x^2\,dx = [x^3]_0^2 = 8$ ← finite.

I is improper when

- Type I : $a \to \infty$ or $b \to \infty$ (or both) (Domain is infinite)
  eg: $\int_0^\infty 3x^2\,dx$.
- Type II : $f(x) \to -\infty, +\infty$ at some point on $[c,b]$ (Range is infinite)
  Eg: $\int_{-2}^{2} \frac{1}{x}\,dx$ * There is vertical asymptote inside $[a,b]$
  ↑ As $x \to 0^-$, $f(x) \to -\infty$; $x \to 0^+$, $f(x) \to +\infty$

Improper Integral as the Area Under the Curve

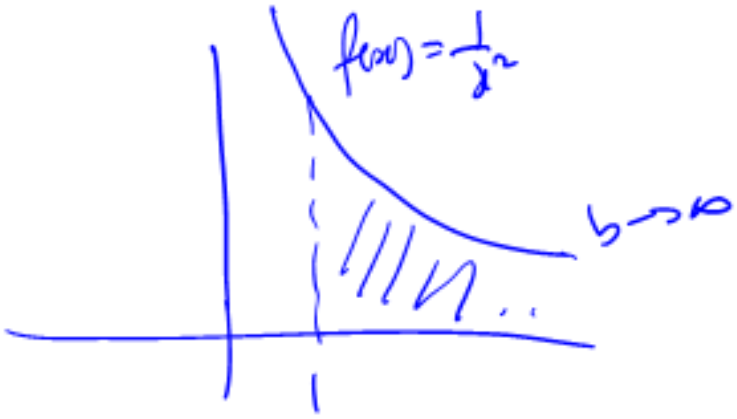
An improper integral of a non-negative function can be related to the area under the curve.



$$\int_1^\infty f(x)\,dx = \int_1^\infty \frac{1}{x}\,dx$$
$$= \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx$$
$$= \lim_{b\to\infty} \left[\ln|x|\right]_1^b$$
$$= \lim_{b\to\infty} \ln b - 0 \to \infty$$
(diverges)

If the area under the curve is finite, then we say the improper integral converges.

Otherwise, we say the improper integral diverges.



$$\int_1^\infty f(x)\,dx = \int_1^\infty \frac{1}{x^2}\,dx$$

$$= \lim_{b\to\infty} \int_1^b \frac{1}{x^2}\,dx$$

$$= \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b$$

$$= \lim_{b\to\infty} \left(-\frac{1}{b}\right) - \left(-\frac{1}{1}\right)$$

$$= 1 \quad \text{(converges)}$$

Type I Improper Integrals

Integrals with infinite limit of integrations are improper integrals of Type I:

- If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx.$$

- If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx.$$

- If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx,$$

where c is any real number.

In each case, if the limit exists and is finite, we say that the improper integral converges and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral diverges.

Example

Is the area under the curve $y = \frac{\ln x}{x^2}$ from $x = 1$ to $x = \infty$ finite? If so, what is its value?

$$\text{Area} = \int_1^\infty \frac{\ln x}{x^2}\,dx \quad \leftarrow \text{cts over } [1, \infty)$$

$$= \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx$$

IBP: Let $u = \ln x$, $du = \frac{1}{x}\,dx$

$\int dv = \int \frac{1}{x^2}\,dx$, $v = -\frac{1}{x}$

$$= \lim_{b\to\infty} \left[-\frac{\ln x}{x} \right]_1^b$$

$$- \lim_{b\to\infty} \int_1^b \left(-\frac{1}{x}\right)\left(\frac{1}{x}\right) dx$$

$\leftarrow -\frac{1}{x^2} = -x^{-2}$

$$= \left[\lim_{b \to \infty} \left(-\frac{\ln b}{b} \right) - 0 \right]$$

L'Hôpital Rule

$$- \lim_{b \to \infty} \left[\frac{1}{x} \right]_1^b$$

$$= \lim_{b \to \infty} \frac{-\frac{1}{b}}{1} - \left(\lim_{b \to \infty} \frac{1}{b} - 1 \right) = 1 \quad \text{(finite)}$$

$\to 0$ $\quad$ $\to 0$

Example

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ → π $\frac{1}{1+x^2}$ is cts over $(-\infty, \infty)$

FTC applies.

$$= \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{\infty} \frac{dx}{1+x^2}$$

$$= \lim_{a\to-\infty} \int_{a}^{0} \frac{dx}{1+x^2} + \lim_{b\to\infty} \int_{0}^{b} \frac{dx}{1+x^2}$$

$$= \lim_{a\to-\infty} [\arctan x]_a^0 + \lim_{b\to\infty} [\arctan x]_0^b$$

$$= \left(0 - \lim_{a\to-\infty} \arctan a\right) + \left(\lim_{b\to\infty} \arctan b - 0\right)$$

$\to -\pi/2$ $\to \pi/2$

$$= -\left(-\tfrac{\pi}{2}\right) + \tfrac{\pi}{2} = \pi$$ (The improper integral converges to the value of π)

Example

For what values of p does the integral $\int_1^\infty \frac{dx}{x^p}$ converge? When the integral converge, what is its value?

For $p=1$, we have $\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx$

$= \lim_{b\to\infty} [\ln x]_1^b$

$= \lim_{b\to\infty} \ln b - \ln 1$

$= \infty$

For $p \neq 1$, we have $\int_1^\infty \frac{1}{x^p}\,dx$.

$$= \lim_{b\to\infty} \int_1^b x^{-p}\,dx$$

$$= \lim_{b\to\infty} \left[\frac{x^{-p+1}}{-p+1}\right]_1^b$$

$$= \lim_{b\to\infty} \frac{b^{-p+1}}{-p+1} - \left(\frac{1}{-p+1}\right)$$

$$= \frac{1}{-p+1} \lim_{b\to\infty} \frac{1}{b^{p-1}} - \left(\frac{1}{-p+1}\right)$$

If $p > 1$, $\frac{1}{b^{p-1}} = \frac{1}{b^c}$ for some $c > 0$

$$\lim_{b \to \infty} \frac{1}{b^c} = 0$$

If $p < 1$, $\frac{1}{b^{p-1}} = \frac{1}{b^d}$ for some $c < 0$

$$= b^{-d} = b^e \text{ for some } e > 0$$

$$\lim_{b \to \infty} b^e = \infty$$

So, $$\int_1^\infty \frac{1}{x^p}\,dx = \begin{cases} -\frac{1}{1-p} = \frac{1}{p-1}, & \text{if } p>1 \\ \\ \infty, & \text{if } p<1 \end{cases}$$

p-integral (p-test)

The improper integral $\int_1^\infty \frac{1}{x^p}\,dx$

- converges if $p>1$
- diverges if $p \le 1$

Type II Improper Integrals

Integrals of functions that become infinite at a point within the interval of integration are improper integrals of Type II:

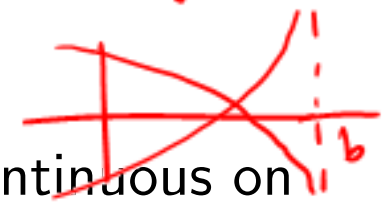
- If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{c \to a^+} \int_c^b f(x)\,dx.$$



- If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{c \to b^-} \int_a^c f(x)\,dx.$$



- If $f(x)$ discontinuous at c where $a < c < b$, and is continuous on $[a, c) \cup (c, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

In each case, if the limit exists and is finite, we say that the improper integral converges and that the limit is the value of the improper integral. If the limit fails to exist, the improper integral diverges.

Example

Investigate the convergence of $\int_0^1 \frac{1}{1-x}\,dx$.

$\frac{1}{1-x}$ ← cts on $[0,1)$ not cts at $x=1$

$$\int_0^1 \frac{1}{1-x}\,dx = \lim_{b\to 1^-} \int_0^b \frac{1}{1-x}\,dx$$

$\int_0^b \frac{1}{1-x}\,dx$ ← cts on $[a,b]$ FTC applies.

$$= \lim_{b\to 1^-} \left[-\ln|1-x|\right]_0^b$$

As $b\to 1^-$, $|1-b|\to 0^+$, $\ln|1-b|\to -\infty$, $-\ln|1-b|\to \infty$

$$= \lim_{b\to 1^-} \left(-\ln|1-b|\right) - 0$$

$$= \infty$$

So, the integral diverges

Example

Evaluate $\displaystyle\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

$f(x) = \frac{1}{(x-1)^{2/3}}$ is cts on $[0,1) \cup (1,3]$

not cts at $x=1$

$$= \int_0^1 \frac{dx}{(x-1)^{2/3}} + \int_1^3 \frac{dx}{(x-1)^{2/3}}$$

$$= \lim_{a \to 1^-} \int_0^a \frac{dx}{(x-1)^{2/3}} + \lim_{b \to 1^+} \int_b^3 \frac{dx}{(x-1)^{2/3}}$$

$$= \lim_{a \to 1^-} \left[\frac{(x-1)^{1/3}}{1/3} \right]_0^a + \lim_{b \to 1^+} \left[\frac{(x-1)^{1/3}}{1/3} \right]_b^3$$

$$= \left[\lim_{a \to 1^-} 3(a-1)^{1/3} - (-3) \right] + \left[3(2)^{1/3} - \lim_{b \to 1^+} 3(b-1)^{1/3} \right]$$

($\lim_{a \to 1^-} 3(a-1)^{1/3} \to 0$, $\lim_{b \to 1^+} 3(b-1)^{1/3} \to 0$)

$$= 3 + 3(2)^{1/3} \quad \text{(converges)}$$

Example

Evaluate $\int_2^\infty \frac{x+3}{(x-1)(x^2+1)}\,dx$.

→ not cts at $x=1$ but it is cts over $[2,\infty)$

No need to split the integral.

Let $\frac{x+3}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$

$\Rightarrow \quad x+3 = A(x^2+1) + (Bx+C)(x-1)$

$= (A+B)x^2 + (C-B)x + (A-C)$

Comparing the coefficients:

$$\left.\begin{aligned} A+B&=0\\ C-B&=1\\ A-C&=3 \end{aligned}\right\} \Rightarrow A=2,\ B=-1,\ C=-1$$

$$\int_2^\infty \frac{x+3}{(x-1)(x^2+1)}\,dx = \int_2^\infty \frac{2}{x-1} + \frac{-2x-1}{x^2+1}\,dx$$

$$= \lim_{b\to\infty} \int_2^b \frac{2}{x-1}\,dx + \lim_{b\to\infty} \int_2^b \frac{-2x-1}{x^2+1}\,dx$$

$$= \lim_{b\to\infty} \left[2\ln|x-1|\right]_2^b - \lim_{b\to\infty} \int_2^b \frac{2x}{x^2+1}\,dx$$

$$- \lim_{b\to\infty} \int_2^b \frac{1}{x^2+1}\,dx$$

substitution

$$= \lim_{b\to\infty} 2\ln|b-1| - \lim_{b\to\infty} \left[\ln|x^2+1|\right]_2^b$$

$$- \lim_{b\to\infty} \left[\arctan x\right]_2^b$$

$$= \lim_{b\to\infty} 2\ln|b-1| - \lim_{b\to\infty} \ln|b^2+1| + \ln 5$$

$$- \lim_{b\to\infty} \operatorname{arctan} b + \operatorname{arctan} 2$$

$$\to \pi/2$$

$$= \lim_{b\to\infty} \left(\ln \frac{(b-1)^2}{b^2+1}\right) + \ln 5 - \frac{\pi}{2} + \operatorname{arctan} 2$$

$$= \ln \lim_{b\to\infty} \frac{(b-1)^2}{b^2+1} + \ln 5 - \pi/2 + \operatorname{arctan} 2$$

$$= \ln \left(\lim_{b\to\infty} \frac{2(b-1)}{2b}\right) + \ln 5 - \pi/2 + \operatorname{arctan} 2$$

$$= \ln \left(\lim_{b\to\infty} 1\right) + \ln 5 - \pi/2 + \operatorname{arctan} 2$$

$$= \ln 5 - \pi/2 + \operatorname{arctan} 2.$$

Example

Evaluate $\int_{-1}^{2} \frac{1}{x^2}\,dx$ $= \left[\frac{-1}{x}\right]_{-1}^{2} = \frac{-3}{2}$ ✗ Not correct

~ ← not cts at $x=0$

* Always check the continuity of the function at the interval !!!

$$\int_{-1}^{2} \frac{1}{x^2}\,dx = \int_{-1}^{0} \frac{1}{x^2}\,dx + \int_{0}^{2} \frac{1}{x^2}\,dx$$

$$= \lim_{a\to 0^-} \int_{-1}^{a} \frac{1}{x^2}\,dx + \lim_{b\to 0^+} \int_{b}^{2} \frac{1}{x^2}\,dx$$

$$= \lim_{a\to 0^-} \left[-\frac{1}{x}\right]_{-1}^{a} + \lim_{b\to 0^+} \left[-\frac{1}{x}\right]_{b}^{2}$$

→ diverge → diverge

So $\int_{-1}^{2} \frac{1}{x^2}\,dx$ diverges.

Tests for Convergence and Divergence

Sometimes, we may not be able to evaluate the improper integral directly.

However, we may still be able to determine whether the integral converges or diverges by comparing it one integral whose convergence properties are known.

Two comparison test:

- Direct Comparison Test
- Limit Comparison Test

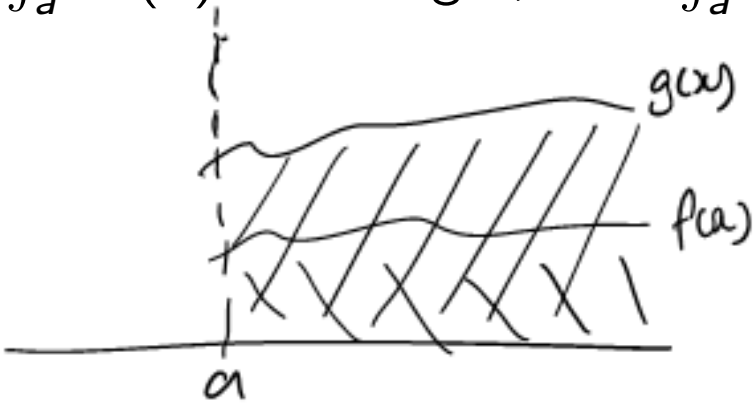
These tests only tell us if the integral converges or diverges. To find (or estimate) its value, if converges, we need to use other (numerical) methods to approximate the value.

In many cases, it is helpful to compare your given integral with that of a p-integral.

Direct Comparison Test

Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then

1. if $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$ also converges.
2. if $\int_a^\infty f(x)\,dx$ diverges, then $\int_a^\infty g(x)\,dx$ also diverges.



$\int_a^\infty g(x)\,dx$ = Area under the curve $g(x)$

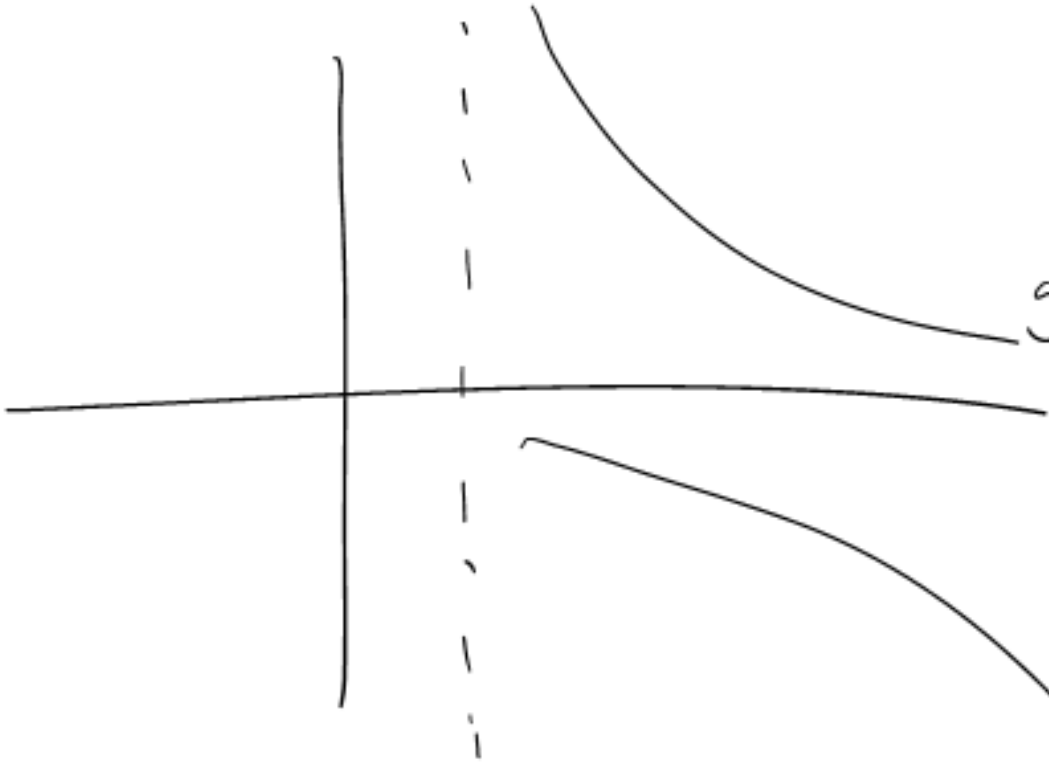
$\int_a^\infty f(x)\,dx$ = Area under the curve $f(x)$

$\int_a^\infty f(x)\,dx$ = Area under $f(x)$ $\leq$ Area under $g(x)$ = $\int_a^\infty g(x)\,dx$

① Area under $g(x)$ is finite $\Rightarrow$ Area under $f(x)$ is finite.

② Area under $f(x)$ is infinite $\Rightarrow$ Area under $g(x)$ is infinite.

It is important to check that both $f(x)$ and $g(x)$ are nonnegative for $x \geq a$



$g(x) = \frac{1}{x^2} \Rightarrow$ converges.
(refer to previous example)

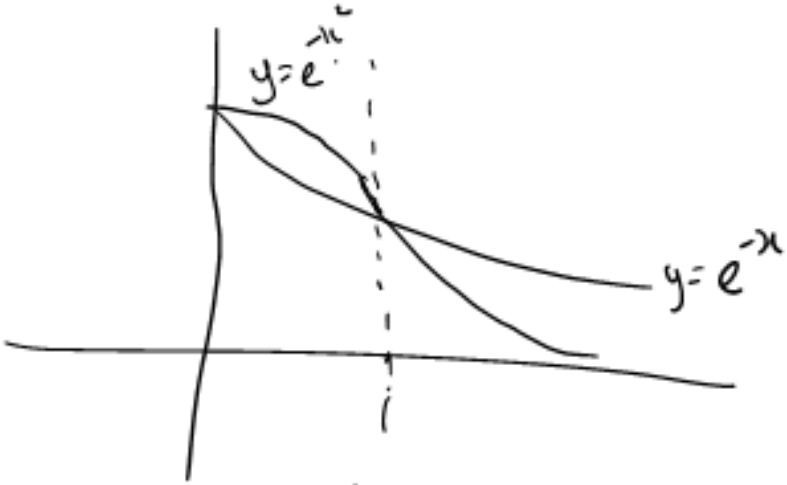
$\int_a^\infty f(x)\,dx$ diverges.

Alternatively, $f(x) = -\frac{1}{x}$

$f(x)$

Direct Comparison Test

Show $\int_1^\infty e^{-x^2}\,dx$ converges



For $x \geq 1$

we have $0 \leq e^{-x^2} \leq e^{-x}$

So, $0 \leq \int_1^\infty e^{-x^2}\,dx \leq \int_1^\infty e^{-x}\,dx$

$= \lim_{b\to\infty}\left[-e^{-x}\right]_1^b$

$= \lim_{b\to\infty}\left(-e^{-b}\right) - \left(-e^{-1}\right)$

$= 0 + e^{-1} = e^{-1}$

(converges)

So, by the Direct Comparison Test,

$\int_1^\infty e^{-x^2}\,dx$ converges.

(to a value less than or equal to e^{-1})

Direct Comparison Test

$\int_1^\infty \frac{\sin^2 x}{x^2}\,dx$ converges

For $x \geq 1$, $0 \leq \sin^2 x \leq 1$

$$0 = \frac{0}{x^2} \leq \frac{\sin^2 x}{x^2} \leq \frac{1}{x^2}$$

So, $\int_1^\infty \frac{\sin^2 x}{x^2}\,dx \leq \int_1^\infty \frac{1}{x^2}\,dx$

We know that $\int_1^\infty \frac{1}{x^2}\,dx$ converges (p-integral with $p = 2 > 1$)

So, by Direct Comparison Test, $\int_1^\infty \frac{\sin^2 x}{x^2}\,dx$ converges.

Direct Comparison Test

$$\int_1^\infty \frac{1}{\sqrt{x^2 - 0.1}}\, dx \text{ diverges}$$

$\frac{1}{n} < \frac{1}{m}$ if $n > m$ and eg: $\frac{1}{3} < \frac{1}{2}$

$\frac{1}{n} > \frac{1}{m}$ if $n < m$ eg: $\frac{1}{3} > \frac{1}{4}$

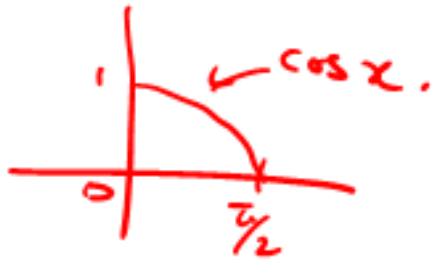
For $x \geq 1$, $\dfrac{1}{\sqrt{x^2 - 0.1}} \geqslant \dfrac{1}{\sqrt{x^2}} = \dfrac{1}{x} \geqslant 0$

We know that $\int_1^\infty \frac{1}{x}\, dx$ diverges (p-integral with $p=1$)

So, by Direct Comparison Test, $\int_1^\infty \frac{1}{\sqrt{x^2-0.1}}\, dx$ diverges.

Direct Comparison Test

$\int_0^{\pi/2} \frac{\cos x}{\sqrt{x}}\, dx$ converges



On $[0, \pi/2]$ $\quad 0 \le \cos x \le 1$

$$0 = \frac{0}{\sqrt{x}} \le \frac{\cos x}{\sqrt{x}} \le \frac{1}{\sqrt{x}}$$

So, $\int_0^{\pi/2} \frac{\cos x}{\sqrt{x}}\, dx \le \int_0^{\pi/2} \frac{1}{\sqrt{x}}\, dx$

$$= \lim_{a \to 0^+} \int_a^{\pi/2} \frac{1}{\sqrt{x}}\, dx$$

Hence, by Direct Comparison Test, $\int_0^{\pi/2} \frac{\cos x}{\sqrt{x}}\, dx$

$$= \lim_{a \to 0^+} \left[2\sqrt{x}\right]_a^{\pi/2} = 2\sqrt{\pi/2} - \lim_{a \to 0^+} 2\sqrt{a}$$

$\to 0$

$= 2\sqrt{\pi/2}$ (converges)

converges.

Direct Comparison Test

Do not use the Direct Comparison Test to test $\int_a^\infty f(x)\,dx$ when

(a) $f(x) \leq g(x)$ and $\int_a^\infty g(x)\,dx$ diverges, or

(b) $g(x) \leq f(x)$ and $\int_a^\infty g(x)\,dx$ converges.

For example, to determine if $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ we cannot use Direct Comparison Test to compare it with $\int_1^\infty \frac{1}{x}$ even though we know $\frac{1-e^{-x}}{x} \leq \frac{1}{x}$. This is because the integral $\int_1^\infty \frac{1}{x}$ diverges, and we cannot say anything about $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$. Instead, we can use the Limit Comparison Test.

Limit Comparison Test

If the positive functions f and g are continuous on $[a, \infty)$, and if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L,\ 0 < L < \infty,$$

finite limit.

then

$$\int_a^\infty f(x)\,dx \text{ and } \int_a^\infty g(x)\,dx$$

either both converge or both diverge.

By the definition of limit $\lim_{x\to\infty} \frac{f(x)}{g(x)} = L$

$\forall \varepsilon > 0$, there is a $M \in \mathbb{R}$ s.t. if $x > M$,

then $\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon$

Consider $0 < \varepsilon < L$, we have

$$\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon$$

$$\Rightarrow \quad -\varepsilon < \frac{f(x)}{g(x)} - L < \varepsilon$$

$$\Rightarrow \quad 0 < L - \varepsilon < \frac{f(x)}{g(x)} < L + \varepsilon$$

$$\Rightarrow \quad 0 < g(x)(L-\varepsilon) < f(x) < g(x)(L+\varepsilon) \quad \text{for } x > M$$

general idea : $\int_M^\infty g(x)\,dx$ diverges $\Rightarrow (L-\varepsilon)\int_M^\infty g(x)\,dx$ div. $\Rightarrow \int_M^\infty f(x)\,dx$ div.

$\int_M^\infty g(x)\,dx$ conv. $\Rightarrow (L+\varepsilon)\int_M^\infty g(x)\,dx$ conv. $\Rightarrow \int_M^\infty f(x)\,dx$ conv.

$\int_M^\infty f(x)\,dx$ conv. $\Rightarrow \frac{1}{L-\varepsilon}\int_M^\infty f(x)\,dx$ conv. $\Rightarrow \int_M^\infty g(x)\,dx$ conv.

$\int_M^\infty f(x)\,dx$ div. $\Rightarrow \frac{1}{L+\varepsilon}\int_M^\infty f(x)\,dx$ div. $\Rightarrow \int_M^\infty g(x)\,dx$ div.

Limit Comparison Test

Show that

$$\int_1^\infty \frac{dx}{1+x^2}$$

converges by comparison with $\int_1^\infty (1/x^2)\,dx$. Find and compare the values of two integrals.

Take $f(x) = \frac{1}{1+x^2}$ and $g(x) = \frac{1}{x^2}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\frac{1}{1+x^2}}{\frac{1}{x^2}} = \lim_{x\to\infty} \frac{x^2}{1+x^2} = 1$$

We know that $\int_1^\infty \frac{1}{x^2}\,dx$ converges (p-integral with $p=2>1$)

So, by Limit Comparison Test, $\int_1^\infty \frac{dx}{1+x^2}$ also converges.

* Again, this only tell you that $\int_1^\infty \frac{dx}{1+x^2}$ converges, it does not tell you its value.

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b = \lim_{b\to\infty} -\frac{1}{b} - \left(-\frac{1}{1}\right)$$

$$= 1$$

$$\int_1^\infty \frac{dx}{1+x^2} = \lim_{b\to\infty}\left[\arctan x\right]_1^b = \lim_{b\to\infty} \arctan b - \arctan 1$$

$$= \pi/2 - \pi/4 = \pi/4$$

Limit Comparison Test

Investigate the convergence of $\displaystyle\int_1^{\infty} \frac{1-e^{-x}}{x}\,dx$.

To choose $g(x)$

(1) We can easily know that $g(x)$ conv or div.
(at least easier than $f(x)$)

(2) $\lim_{x\to\infty} \frac{f(x)}{g(x)} \Big/ \lim_{x\to\infty} \frac{g(x)}{f(x)}$ can be easily evaluated.

Take $f(x) = \dfrac{1-e^{-x}}{x}$, $g(x) = \dfrac{1}{x}$

$$\lim_{x\to\infty} \frac{\frac{1-e^{-x}}{x}}{1/x} = \lim_{x\to\infty} 1-e^{-x}$$

$$= 1$$

We know that $\int_1^{\infty} \frac{1}{x}\,dx$ diverges (p-integral with $p=1$)

So, by Limit Comparison Test, $\int_1^{\infty} \frac{1-e^{-x}}{x}\,dx$ diverges.

Limit Comparison Test

Investigate the convergence of $\displaystyle\int_2^{\infty} \frac{\sqrt{3x^2+\sin x}}{x^4-2}\,dx$.

Take $f(x) = \dfrac{\sqrt{3x^2+\sin x}}{x^4-2}$

As $x \to \infty$, $\sqrt{3x^2+\sin x} \longrightarrow \sqrt{3x^2}$ and $x^4-2 \to x^4$

So $f(x) \simeq \dfrac{\sqrt{3x^2}}{x^4} = \dfrac{\sqrt{3}}{x^3}$

Take $g(x) = \dfrac{\sqrt{3}}{x^3}$

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \frac{\lim_{x\to\infty} f(x)}{\lim_{x\to\infty} g(x)} = \frac{\lim_{x\to\infty} \frac{\sqrt{3}}{x^3}}{\lim_{x\to\infty} \frac{\sqrt{3}}{x^3}} = 1$$

Since $\int_1^{\infty} \frac{1}{x^3}\,dx$ converges (p-integral with $p=3>1$)

By DCT, $\int_2^\infty \frac{1}{x^3}\,dx$ converges.

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$$\Rightarrow \int_2^\infty \frac{\sqrt{3}}{x^3}\,dx \text{ converges.}$$

So, by Limit Comparison Test, $\int_2^\infty \frac{\sqrt{3x^4+\sin x}}{x^4-2}\,dx$

also converges.